SHASHWAT YADAV

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SKILLS SUMMARY

- **Languages:** C++, C#, Java, Python, SQL, JavaScript
- **Core CS & Architecture:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Distributed Systems, System Design, Microservices, gRPC
- **Backend & Infrastructure:** FastAPI, Django, Docker, Redis, Celery, REST APIs, Git, Unit/Integration Testing (95%+ coverage)
- **Databases:** PostgreSQL, MySQL, MongoDB, ChromaDB (VectorDB)
- **AI/ML Integration:** Large Language Models (LLMs), RAG Pipelines, Prompt Engineering, PyTorch, Semantic Search, NLP, GitHub Copilot

EXPERIENCE

AI Engineer Intern *Jan 2026 – May 2026*
Hasprana Health Care Solutions · Bengaluru, India

- Designed and automated healthcare data pipelines using Java and Python, scaling backend throughput to process 10,000+ daily records and reducing manual operational overhead by 30% through automation scripts.
- Architected the production deployment infrastructure for predictive machine learning models, optimizing system integration to improve forecasting accuracy by 15%.
- Integrated Large Language Model (LLM) services into enterprise microservices using RAG and prompt engineering techniques, collaborating within an 8-person Agile squad to deliver customer-facing features across two-week sprints.
- Strengthened system reliability by authoring 40+ unit and integration tests, maintaining over 95% code coverage for all newly developed ingestion modules.
- Contributed to mobile application development, building a RAG-based virtual avatar AI chatbot using prompt engineering to resolve user queries, alongside features such as live location sharing, instant calling, and ambulance tracking.

PROJECTS

- **VectorDiff – RAG Delta-Sync Engine | Github**
  Tech : Python, ChromaDB, Typer, Rich, Multithreading, gRPC
  - **Designed a** chunk-aware Retrieval-Augmented Generation (RAG) diffing engine, reducing vector embedding API calls and indexing latency by >90% by implementing Hirschberg's space- improved LCS algorithm from scratch.
  - **Architected** a concurrent indexing pipeline (ThreadPoolExecutor + ChromaDB) with zero race conditions, and built a custom Leaky Bucket rate limiter to eliminate rate-limit exhaustion during high-volume syncs.
- **AeroCache — A Cache Memory System | Github**
  Tech :Java · Consistent Hashing · LRU/LFU Eviction · Multi-Node Systems
  - **Created** a fault-tolerant in-memory caching system in Java, achieving O(1) latency for data retrieval by implementing a custom thread-safe HashMap and an intrusive Doubly-Linked List from scratch.
  - **Architected** a master-slave replication pipeline featuring automated heartbeat failover and monotonic epoch split-brain prevention, maintaining strict state consistency and high availability across multi-node clusters.
- **NexusSearch — Sharded Vector Search Engine | Github**
  Tech : Python, gRPC, SQLite, MapReduce, KD-Trees
  - **Constructed** a decentralized search engine utilizing a custom gRPC-based MapReduce framework, scaling concurrent query fan-outs across multiple SQLite-sharded worker nodes to maintain sub-50ms search latency.
  - **Engineered** a custom KD-Tree for high-performance vector similarity search, accelerating query execution by implementing hyperplane-distance pruning to rapidly bypass low-probability spatial regions.
- **Sandboxed Algorithmic Judge & AI Code Reviewer | Github**
  Tech : Python, Docker, Redis, Celery, PostgreSQL, Ollama
  - **Engineered** a highly secure,dynamic code execution platform to evaluate untrusted C++ and Python submissions, leveraging the Docker SDK to isolate processes in sibling containers with fully dropped Linux privileges.
  - **Architected** an asynchronous, fault-tolerant job processing pipeline utilizing FastAPI, Redis as a message broker, and Celery workers, seamlessly sustaining high-throughput concurrent code evaluations without blocking the API gateway.
  - **Integrated** a local Large Language Model (Qwen 2.5 Coder) via Ollama to perform automated static analysis, instantly generating actionable architectural feedback and time/space complexity reviews for user submissions.

EDUCATION

B.E. — Artificial Intelligence & Machine Learning **CGPA: 9.15 /10**
Dayananda Sagar Academy of Technology & Management, Bengaluru *Oct 2023 –June 2027**
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Distributed Systems, System Design, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Deep Learning, Image Processing, Computer Vision, Natural Language Processing, Reinforcement Learning.

ACHIEVEMENTS & PUBLICATIONS

- **BMS Hackathon — 1st Place:** Medical Report Simplifier (LLM + Django); outperformed all 70 competing teams.
- **IEEE Publication — NQComp 2026:** "A Review on AI-Enabled Wildlife Preservation and Management System" — IEEE International Conference on Next-Gen Quantum and Advanced Computing.
- **Smart India Hackathon 2025 — Institutional Nominee:** Selected from 100+ teams for innovative AI/ML technical solution.

CERTIFICATIONS

- Agentic AI Executive — ServiceNow
- Certified System Administrator — ServiceNow
- Machine Learning Fundamentals — Infosys Springboard
- Introduction to NLP — Infosys Springboard